

มหาวิทยาลัยมหิดล
Mahidol University



ITCS215
Introduction to Interactive Multimedia Technology

Semester 1/2026

Image Segmentation

Assoc. Prof. Dr. Worapan Kusakunniran

Image Segmentation

- Examples (Interactive systems)



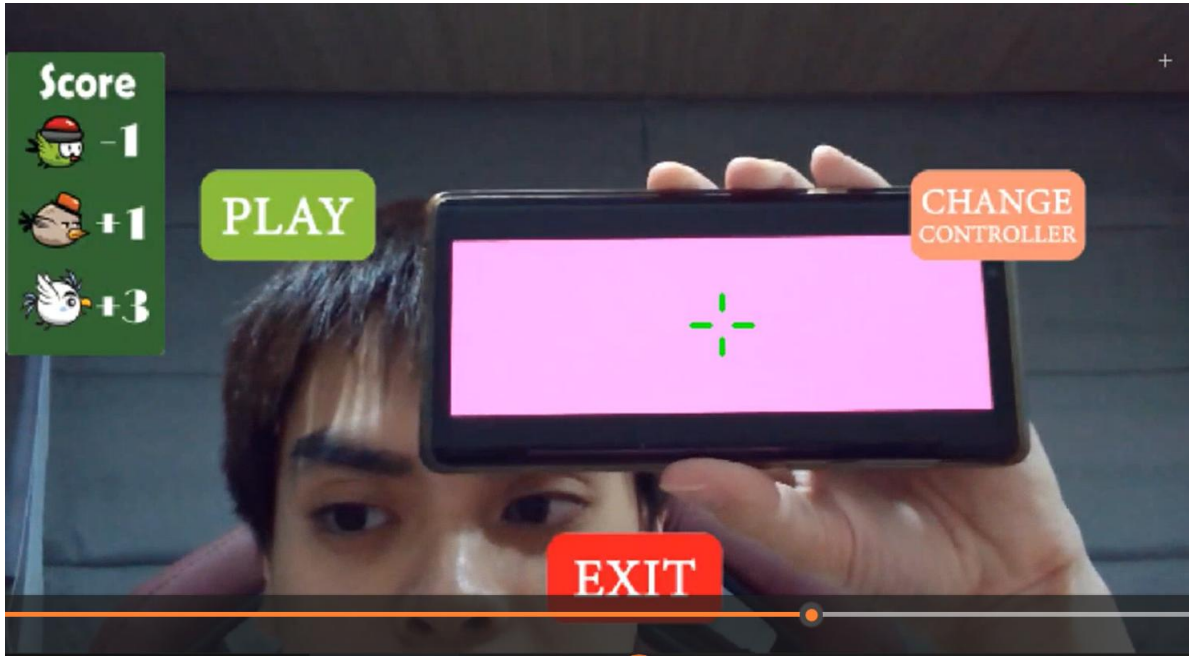
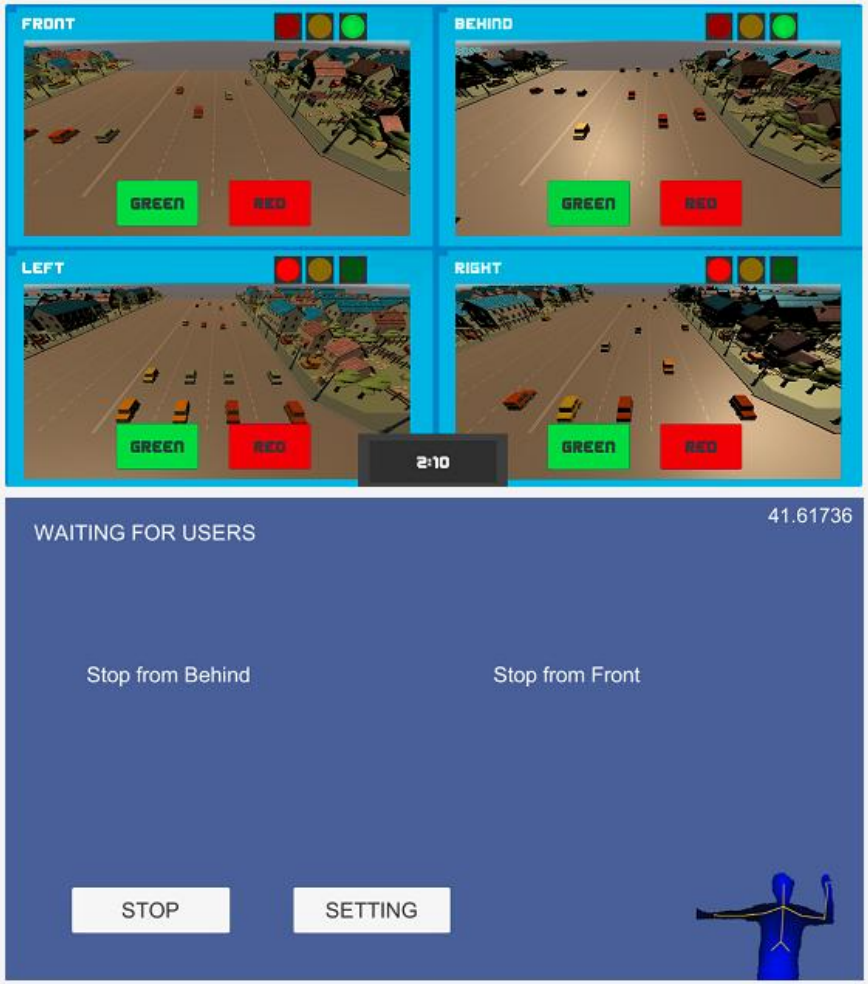
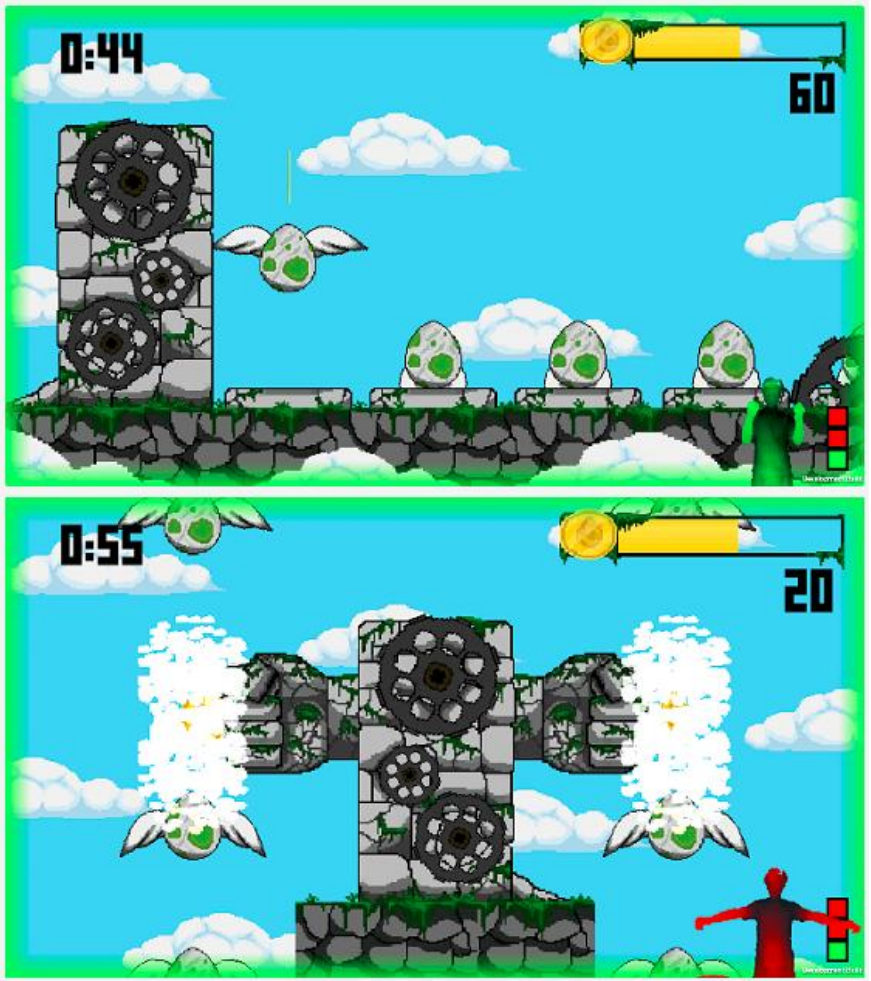


Image Segmentation

- Examples (Interactive systems)

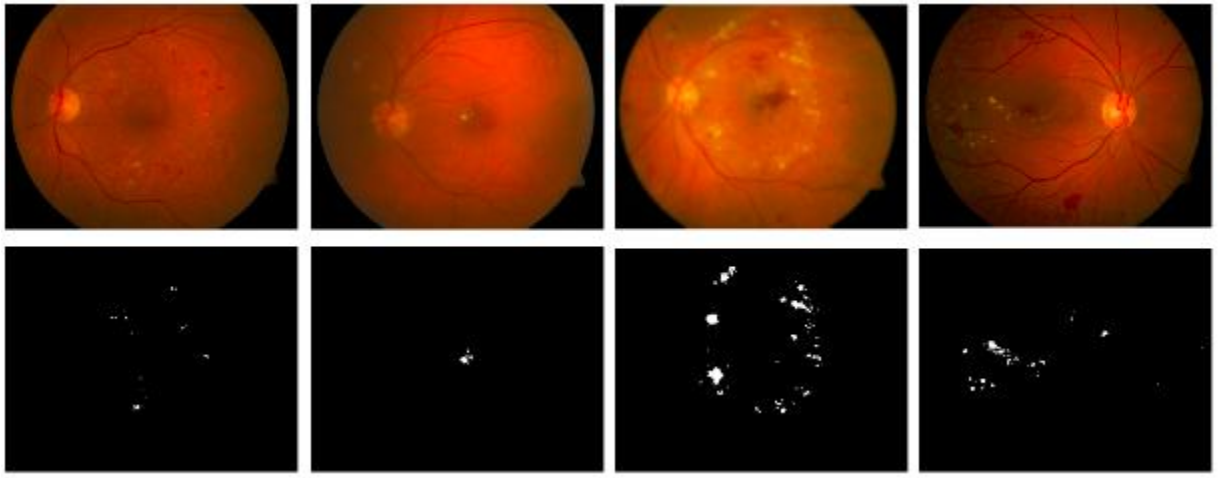


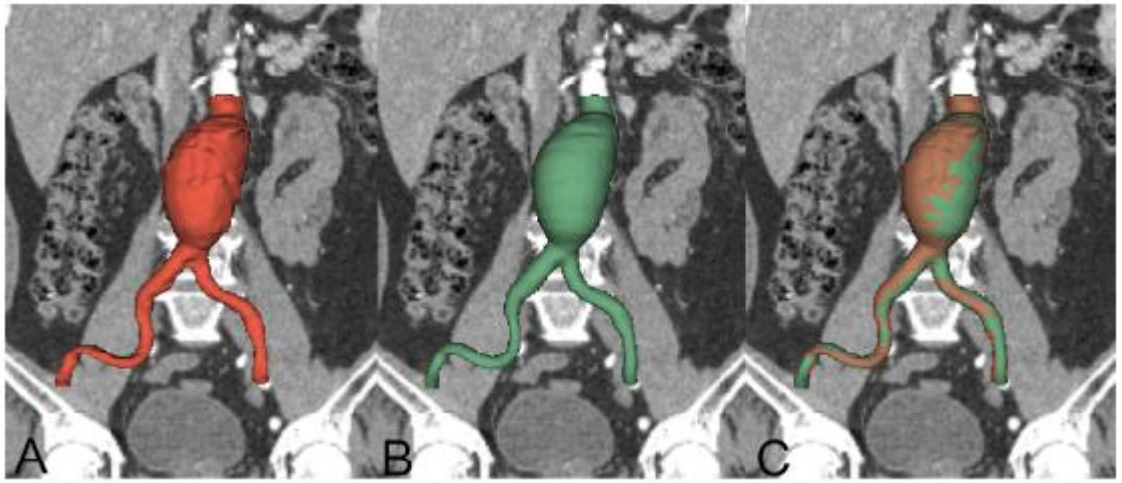


https://sites.google.com/a/mahidol.edu/worapan-kusakunniran/cluster2_vision-based-human-interaction

Image Segmentation

- Examples (Medical image processing)

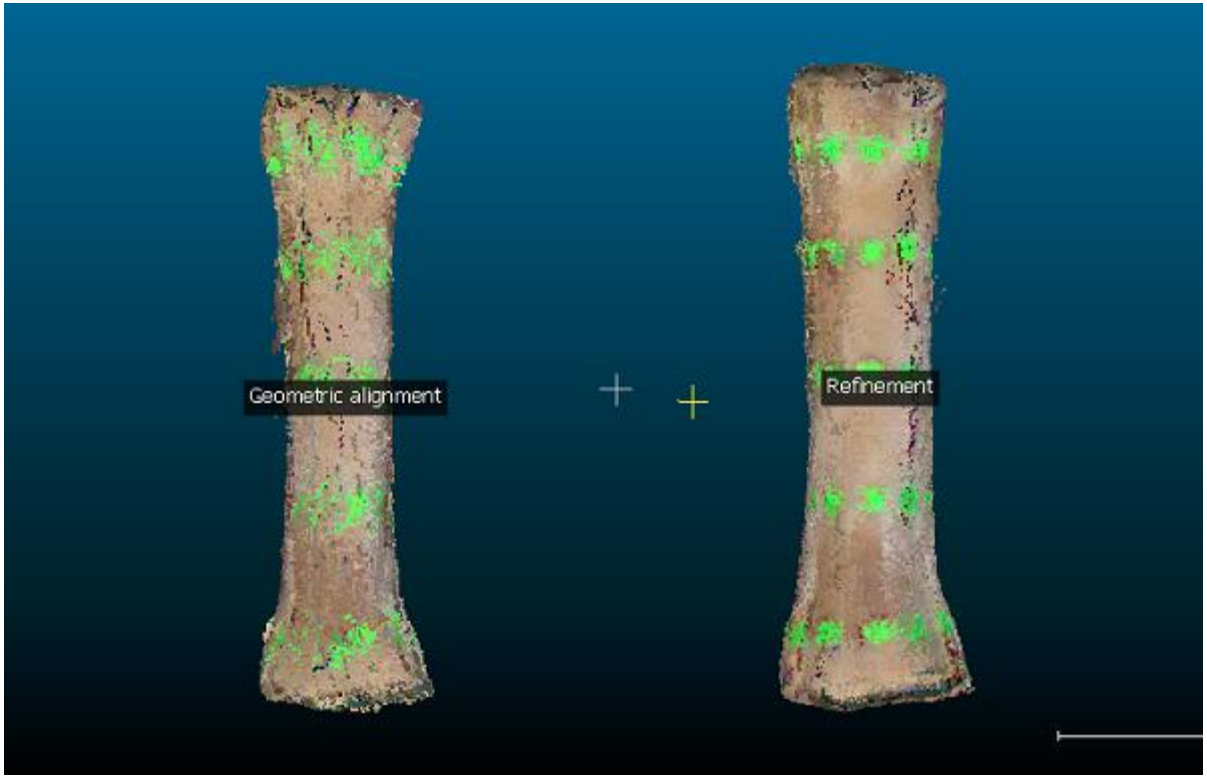




https://sites.google.com/a/mahidol.edu/worapan-kusakunniran/cluster3_medical-image-processing

Image Segmentation

- Examples (3D Reconstruction)



https://sites.google.com/a/mahidol.edu/worapan-kusakunniran/cluster6_ai-cv-based-application

Image Segmentation

- Examples (Biometrics)



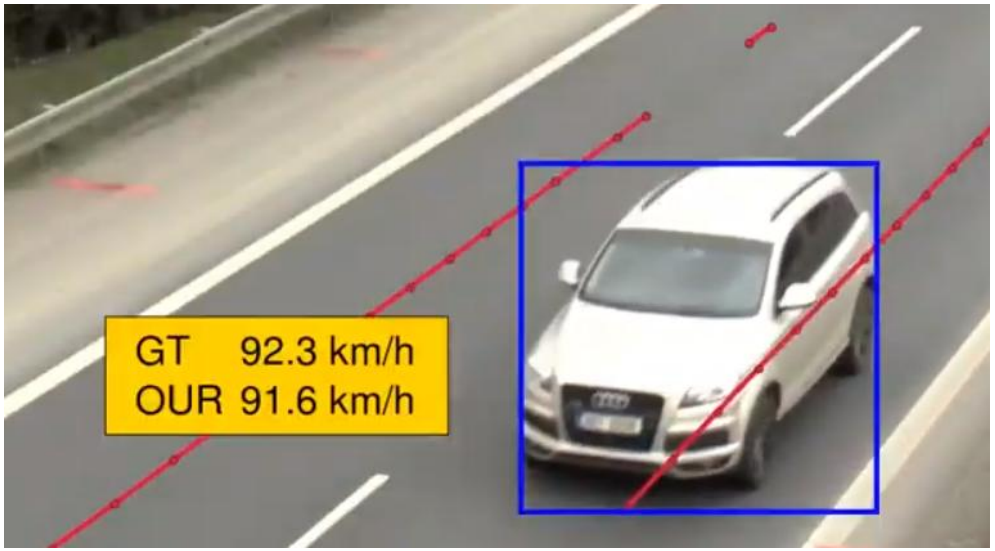


CASIA gait database B

Image Segmentation

- Examples (Smart surveillance)





https://www.kaggle.com/datasets?tagids=14216

https://medusa.fit.vutbr.cz/traffic/traffic-surveillance-camera-calibration-by-3d-model-bounding-box-alignment-for-accurate-vehicle-speed-measurement/

Image Segmentation

- Examples (Scene mapping)

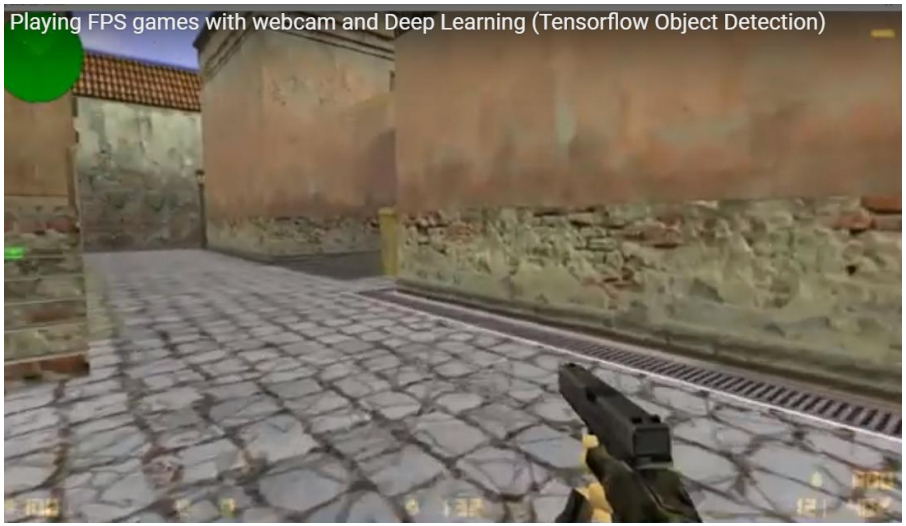


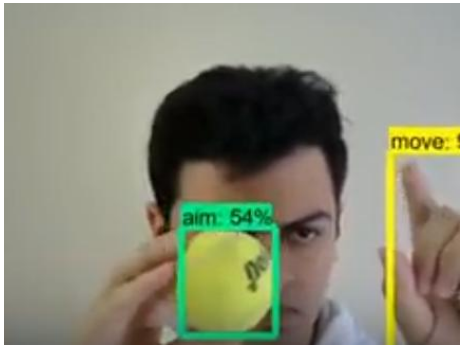
https://www.youtube.com/watch?v=SHickY0NY-Y

Image Segmentation

- Examples (Webcam game)







https://www.youtube.com/watch?v=mUMbgdQ8mOM
https://www.youtube.com/watch?v=V5ARsqnWljY&t=863s

Image Segmentation

- Grey-level image segmentation
  - Conversion from a grey-level image into a bi-level image (i.e., black and white) – foreground/background
  - Extract essential information on an image e.g.
    - The number of objects
    - The positions of objects
    - The shapes of objects
  - Reducing other non-important information on an image



docs.opencv.org

Image Segmentation

- Image thresholding
  - Preset thresholding
  - Mean thresholding
  - Fixed ratio
  - Iterative selection
  - Entropy thresholding
- Iterative graph cut
  - Grabcut
  - Scene mapping

Image Thresholding

- Concept:

All grey levels below the threshold will be classified as black (i.e., foreground), and those above will be white (i.e. background)

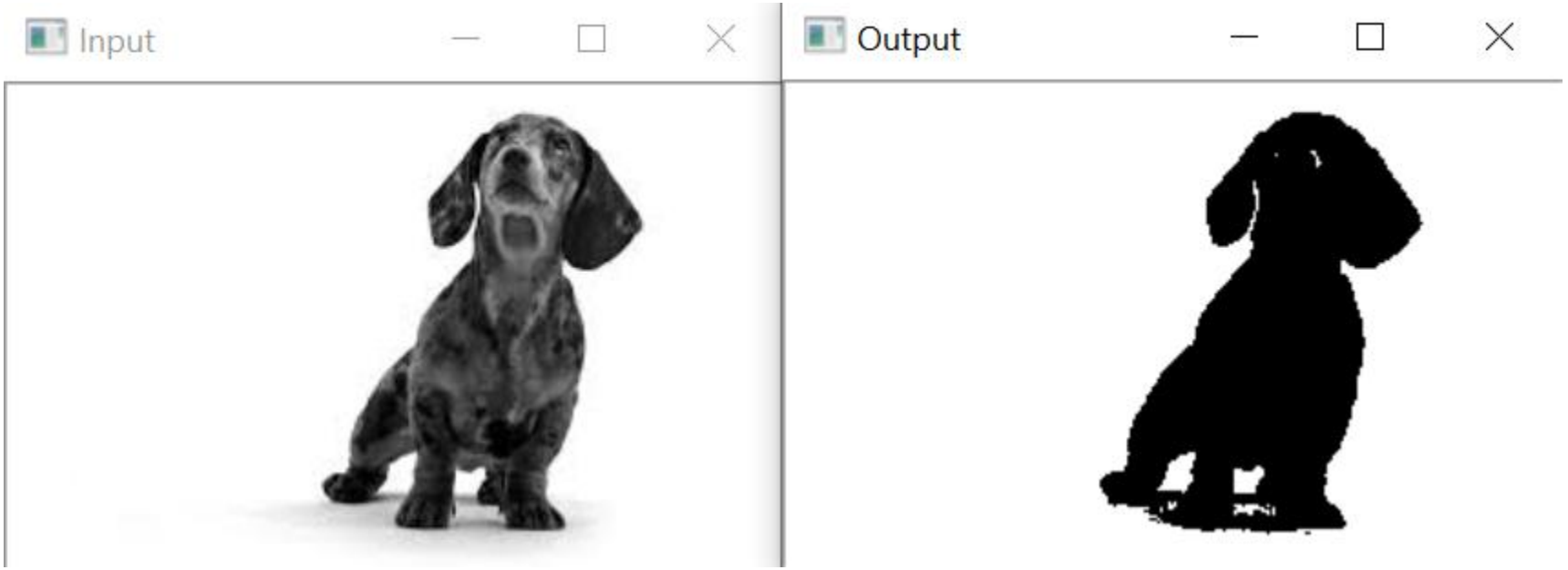


Image Thresholding

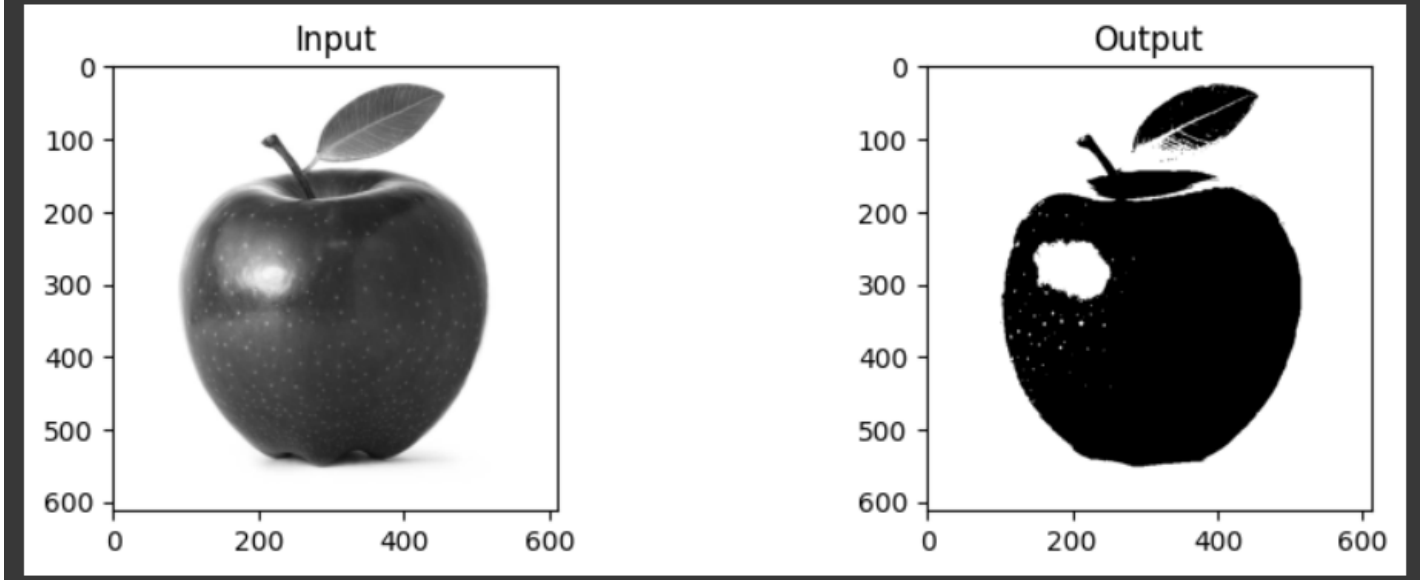
```python
import cv2
img = cv2.imread('/content/drive/MyDrive/Colab Notebooks/3.1_apple.jpg',0)
ret,thresh = cv2.threshold(img,128,255,cv2.THRESH_BINARY)

import matplotlib.pyplot as plt
fig = plt.figure(figsize=(10, 3))
fig.add_subplot(1, 2, 1)
plt.imshow(img, cmap='gray', vmin=0, vmax=255)
plt.title("Input")

fig.add_subplot(1, 2, 2)
plt.imshow(thresh, cmap='gray', vmin=0, vmax=255)
plt.title("Output")

plt.show()
```

How to make the segmentation result better?



Mean Thresholding

```
import cv2
img = cv2.imread('/content/drive/MyDrive/Colab Notebooks/3.1_apple.jpg',0)
means = cv2.mean(img)
print(means[0])
ret,thresh = cv2.threshold(img,means[0],255,cv2.THRESH_BINARY)

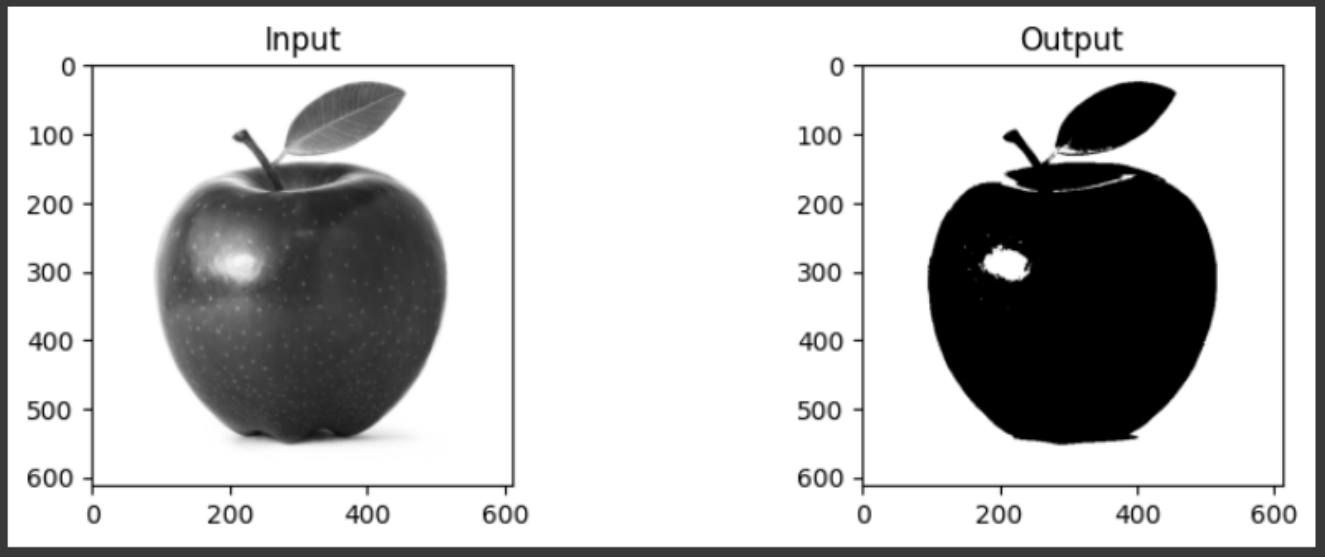
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(10, 3))
fig.add_subplot(1, 2, 1)
plt.imshow(img, cmap='gray', vmin=0, vmax=255)
plt.title("Input")

fig.add_subplot(1, 2, 2)
plt.imshow(thresh, cmap='gray', vmin=0, vmax=255)
plt.title("Output")

plt.show()
```

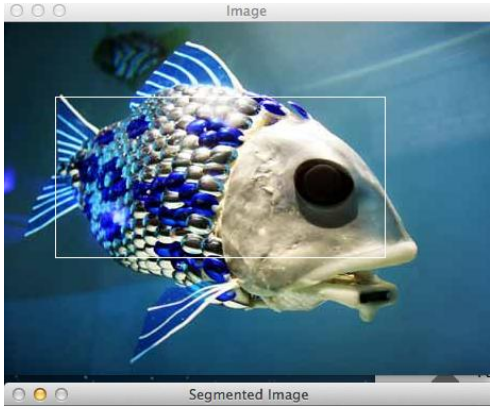
Mean = 182

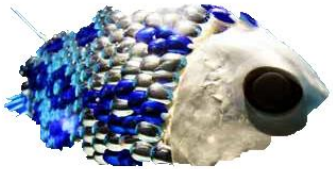
```
182.65090082874107
```



GrabCut Algorithm

- Foreground extraction with minimal user interaction







www.slideshare.net



docs.opencv.org

GrabCut Algorithm (cont.)

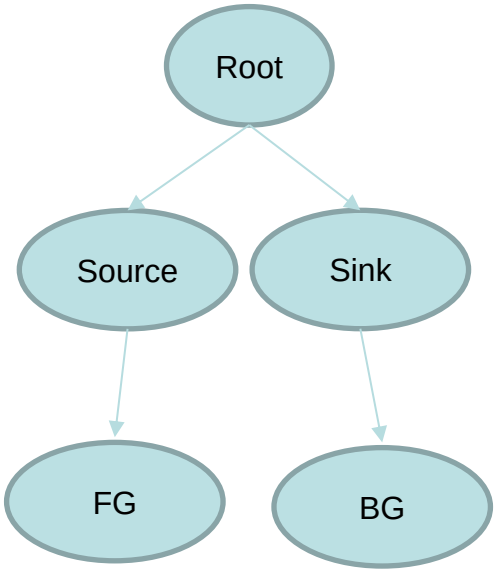
- User inputs the **rectangle**.
  - Everything outside this rectangle will be taken as sure background
  - Everything inside rectangle is unknown

- Computer does an initial labelling depending on the given data. It labels the foreground and background pixels (or it **hard labels**)

- Gaussian Mixture Model(GMM) is used to model the foreground and background

GrabCut Algorithm (cont.)

- Unknown pixels are labelled either probable foreground or probable background depending on its relation with the other hard-labelled pixels in terms of color statistics (i.e., GMM)

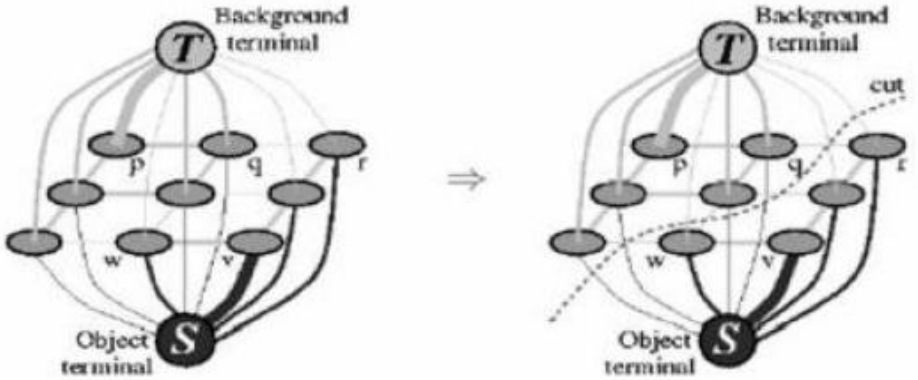


- A graph is built from this pixel distribution
  - (Pixel-nodes) Nodes in the graphs are pixels
  - Additional two nodes are added (Source node and Sink node)
    - Every foreground pixel is connected to Source node
    - Every background pixel is connected to Sink node

GrabCut Algorithm (cont.)

- The weights of edges connecting pixels to source node/end node are defined by the probability of a pixel being foreground/background.

- The weights between the pixels are defined by the edge information or pixel similarity.
  - If there is a large difference in pixel color, the edge between them will get a low weight



(b) Graph. (c) Cut.

GrabCut Algorithm (cont.)

- Then a min-cut algorithm is used to segment the graph
  - It cuts the graph into two separating/disjoint source node and sink node with minimum cost function
  - The cost function is the sum of all weights of the edges that are cut.
  - After the cut, all the pixels connected to Source node become foreground and those connected to Sink node become background

- The process is continued until the classification converges

GrabCut Algorithm (cont.)

- cv2.grabCut(img, mask, rect, bgdModel, fgdModel, iterCount, mode)
  - img – Input 8-bit 3-channel image
  - mask – Input/output 8-bit single-channel mask. 4 possible values
    - **GC_BGD defines an obvious background pixels (0)**
    - **GC_FGD defines an obvious foreground (object) pixel (1)**
    - **GC_PR_BGD defines a possible background pixel (2)**
    - **GC_PR_FGD defines a possible foreground pixel (3)**
  - rect – It is the coordinates of a rectangle which includes the foreground object in the format (x,y,w,h)

https://docs.opencv.org/3.0-beta/modules/imgproc/doc/miscellaneous_transformations.html

GrabCut Algorithm (cont.)

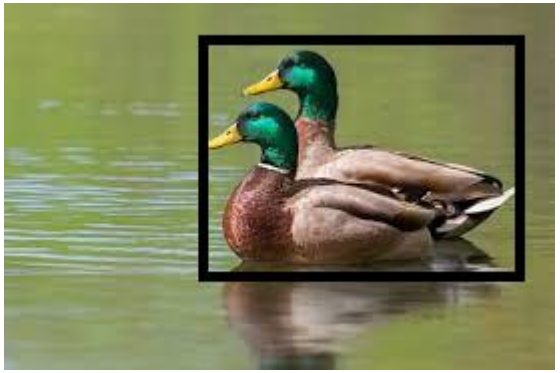
- cv2.grabCut(img, mask, rect, bgdModel, fgdModel, iterCount, mode)
  - bdgModel, fgdModel – Arrays used by the algorithm internally
  - iterCount – Number of iterations the algorithm should run
  - mode – 3 possible modes
    - GC_INIT_WITH_RECT The function initializes the state and the mask using the provided rectangle. After that it runs iterCount iterations of the algorithm.
    - GC_INIT_WITH_MASK The function initializes the state using the provided mask. Note that GC_INIT_WITH_RECT and GC_INIT_WITH_MASK can be combined. Then, all the pixels outside of the ROI are automatically initialized with GC_BGD .
    - GC_EVAL The value means that the algorithm should just resume.

https://docs.opencv.org/3.0-beta/modules/imgproc/doc/miscellaneous_transformations.html

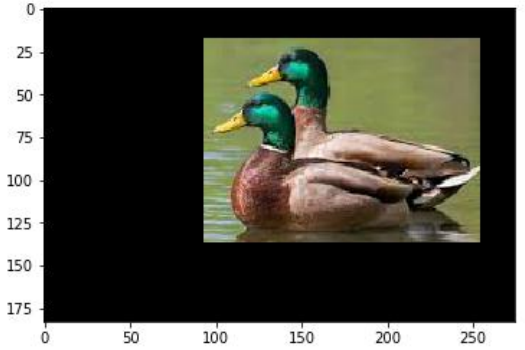
GrabCut Algorithm (cont.)

Act 4.1: GrabCut with initial Rectangle

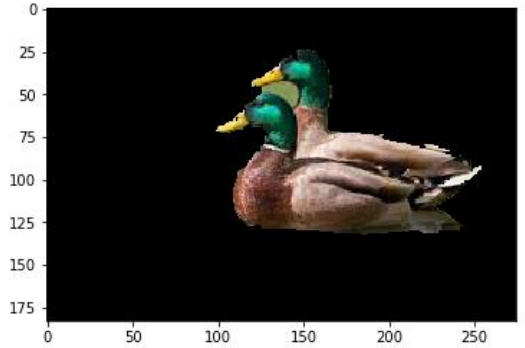




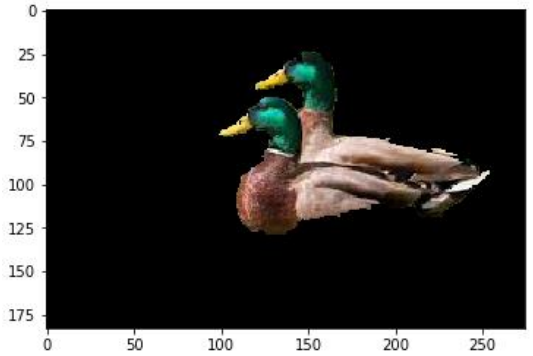
Iteration 0



1



2



GrabCut Algorithm (cont.)

Act 4.1: GrabCut with initial Rectangle

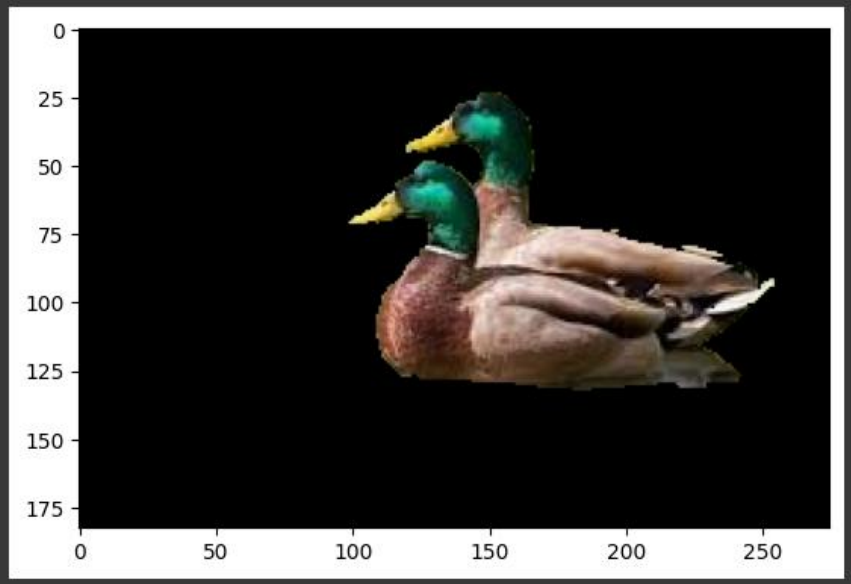
```
import numpy as np
import cv2 as cv
from matplotlib import pyplot as plt

img = cv.imread('/content/drive/MyDrive/Colab Notebooks/duck.jpg')
img = cv.cvtColor(img, cv.COLOR_BGR2RGB)
mask = np.zeros(img.shape[:2],np.uint8)
bgdModel = np.zeros((1,65),np.float64)
fgdModel = np.zeros((1,65),np.float64)
rect = (93,18,162,127)

cv.grabCut(img,mask,rect,bgdModel,fgdModel,2,cv.GC_INIT_WITH_RECT)

mask2 = np.where((mask==2)|(mask==0),0,1).astype('uint8')
img = img*mask2[:,:,np.newaxis]
plt.imshow(img)
plt.show()
```

Change a picture



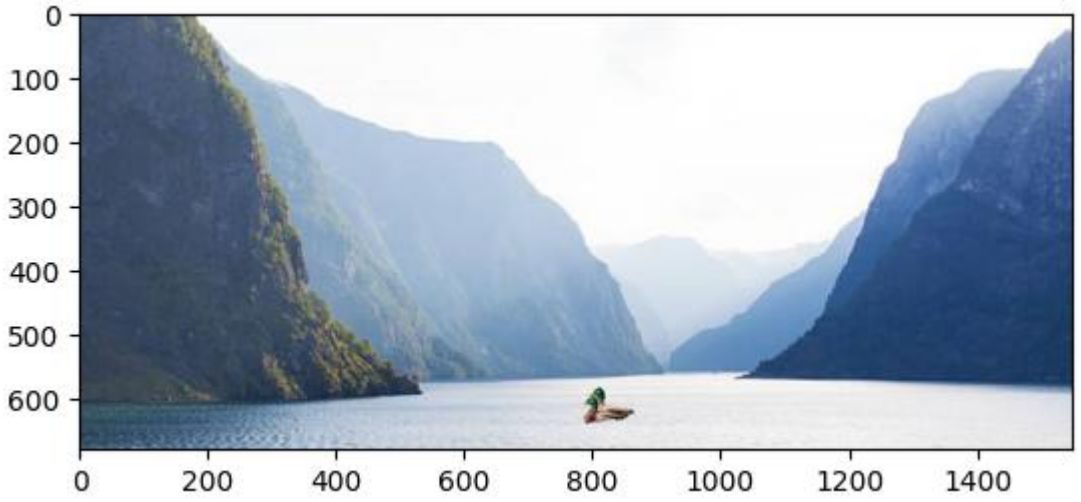
GrabCut Algorithm (cont.)

Act 4.2: Scene Mapping





The mapped Scene must make a sense of a story e.g., two ducks in the lake.



To submit

Group work (DON'T FORGET to put your members' names)
Activity 4.1
Write Python code to perform a GrabCut.

Activity 4.2
Write Python code to read another image and paste the first segmented image onto it at a specific location and scale.